

Robot Framework

Assignment 3: Custom Keywords and Libraries

1. Problem Statement: -

Implement custom keywords to abstract logic, and optionally link external Python libraries to extend capabilities.

2. Objective: -

Learn how to abstract logic into reusable Custom Keywords and integrate Python-based custom scripts into Robot Framework.

3. Tools, Software, and Concepts Used: -

Robot Framework, Python.

4. Source Code / Implementation: -

CustomLibrary.py:-

```
def calculate_discount(price, discount_percent):  
    final_price = float(price) - (float(price) * (float(discount_percent)  
/ 100))  
    return round(final_price, 2)
```

test_automation.robot:-

*** Settings ***

Library SeleniumLibrary

Library CustomLibrary.py

*** Test Cases ***

Verify Custom Keywords And Libraries

[Documentation] Using user-defined keywords and a custom python
library

Print Custom Welcome Message Test Automation User

\${discounted_price}= Calculate Discount 100 20

Should Be Equal As Numbers *\${discounted_price}* 80.0

*** Keywords ***

Print Custom Welcome Message

[Arguments] *\${name}*

Log To Console *\nHello \${name}, welcome to custom keywords!*

5. Output & Screenshots: -

Test Custom Log

Generated
20260923 21:18:55 UTC+05:30
7 seconds ago

REPORT

Statistics

Total Statistics	Total	Pass	Fail	Skip	Elapsed	Pass / Fail / Skip
All Tests	1	1	0	0	00:00:00	

Statistics by Tag	Total	Pass	Fail	Skip	Elapsed	Pass / Fail / Skip
No Tags						

Statistics by Suite	Total	Pass	Fail	Skip	Elapsed	Pass / Fail / Skip
Test Custom	1	1	0	0	00:00:00	

Execution Log

SUITE Test Custom

Full Name: Test Custom
Source: D:\Wipro Selenium\test_custom.robot
Start / End / Elapsed: 20260923 21:18:54.981 / 20260923 21:18:55.232 / 00:00:00.251
Status: 1 test total, 1 passed, 0 failed, 0 skipped

00:00:00.251

TEST Verify Custom Keywords And Libraries

00:00:00.004

Fig.1: Test Custom Log

Test Custom Report

Generated
20260923 21:18:55 UTC+05:30
20 seconds ago

LOG

Summary

Status: All tests passed
Start Time: 20260923 21:18:54.981
End Time: 20260923 21:18:55.232
Elapsed Time: 00:00:00.251
Log File: log.html

Statistics

Total Statistics	Total	Pass	Fail	Skip	Elapsed	Pass / Fail / Skip
All Tests	1	1	0	0	00:00:00	

Statistics by Tag	Total	Pass	Fail	Skip	Elapsed	Pass / Fail / Skip
No Tags						

Statistics by Suite	Total	Pass	Fail	Skip	Elapsed	Pass / Fail / Skip
Test Custom	1	1	0	0	00:00:00	

Details

All Tags Suites Search

Suite:
Test:
Include:
Exclude:

Search Clear Help

Fig.2: Test Custom Report

```

PS D:\Wipro Selenium> robot test_custom.robot
=====
Test Custom
=====
Verify Custom Keywords And Libraries :: Using user-defined keyword...
Hello Test Automation User, welcome to custom keywords!
Verify Custom Keywords And Libraries :: Using user-defined keyword... | PASS |
-----
Test Custom | PASS |
1 test, 1 passed, 0 failed
=====
Output: D:\Wipro Selenium\output.xml
Log:    D:\Wipro Selenium\log.html
Report: D:\Wipro Selenium\report.html

```

Fig.3: Terminal output

6. Result, Observation, and Conclusion: -

- Result: Successfully created a composite Robot Keyword and imported a native Python function as a custom library keyword.
- Observation: Custom keywords help to organize tests into domain-specific language, while Python libraries allow bridging complex backend logic (like calculations or DB queries) into the UI tests.
- Conclusion: Robot Framework's true power lies in its extensibility via custom keyword mapping.